# Freight Rate Prediction

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## 1. The problem

The task is to predict `posted_rate` for 12,000 loads in November and December 2025, using 48,000 labeled loads from January to October 2025. Every load to predict comes after the training data, so this is a forecasting problem. The validation therefore has to test the model on future months.

## 2. Why not a random split?

A random split trains on the same days it tests on, which can make a bad feature look good. `quote_signal` shows this. Its correlation with rate per mile is about +0.99 in some months, about −0.99 in others, and about 0 in August. November and December look like August, so August is the fairest test of this feature.

| LightGBM, MAPE | Random split | Train on other months, test on August |
|---|---|---|
| Without `quote_signal` | 2.98% | 2.06% |
| With `quote_signal` | 2.03% | 4.53% |

The random split says to keep the feature, and the August test says it more than doubles the error. I dropped it.

## 3. How I split the data

Each test window is two months, the same length as November-December. Training always ends before the test window starts, and the test windows don't overlap.

| Fold | Train on | Test on |
|---|---|---|
| Mar-Apr | Jan-Feb | Mar-Apr |
| May-Jun | Jan-Apr | May-Jun |
| Jul-Aug | Jan-Jun | Jul-Aug |
| Sep-Oct | Jan-Aug | Sep-Oct |
| Leave-Aug (extra check, not in the averages) | All months except Aug | Aug |

The corrupted-rate filter and the weight fill are calculated from each fold's training data only. Market values for a date use only loads from that date. Spotter's metric isn't known, so I compare models mainly on MAPE and also track MAE, RMSE, and bias.

## 4. Data problems

| Problem | Size | What I did |
|---|---|---|
| Corrupted rates | 677 rows (1.41%), about 2-6x off | Removed from training only |
| Negative weights | 0.61% of training rows | Took the absolute value |
| Missing weights | 0.62% of training, 1.38% of validation | Filled with the training median |
| Missing `market_index` | 0.78% of training, 2.08% of validation | Filled with that day's average |
| Cities not in training | 8 cities, 12.1% of validation rows | Used coordinates, not city names |

## 5. The model

Two findings from the data shaped the model. First, effects act as percentages: a Reefer costs about 13% more than a Dry Van at any distance, while the dollar gap grows from $68 to $566. Second, rates depend on both the market and a slow drift. January and September have similar `market_index` values, but September rates are about 6% higher.

LightGBM struggles with the second point. It can't predict above the levels it has seen, and with `market_index` added, its Sep-Oct error rises from 2.08% to 4.93%. So I used a ridge regression with splines:

$$\log(\text{rate}) = \text{smooth}(\text{distance, weight, coordinates}) + \text{equipment} + \text{market} + \text{time}$$

The time term stops at the last training date, so the trend isn't projected into the future. Predicting log rate clearly beats predicting dollars with the same design: 2.38% vs 5.30% MAPE.

## 6. Results

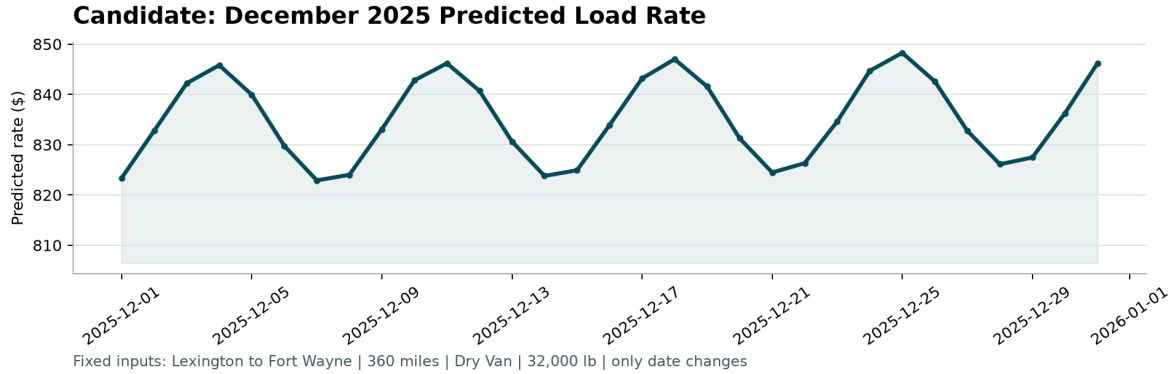
| MAPE % on clean rows | Mean | Worst | Bias | Mar-Apr | May-Jun | Jul-Aug | Sep-Oct | Leave-Aug |
|---|---|---|---|---|---|---|---|---|
| Median baseline | 4.73 | 6.17 | −2.89 | 5.26 | 6.17 | 3.80 | 3.68 | 3.71 |
| LightGBM | 3.76 | 5.87 | −2.90 | 4.67 | 5.87 | 2.42 | 2.08 | 2.06 |
| **Ridge (final)** | **2.38** | **3.00** | **−0.45** | 2.21 | 2.07 | 3.00 | 2.24 | 1.90 |

Mean, worst fold, and bias cover the four forward folds. On a random split the same ridge scores 1.66%, which is roughly the noise floor. With eight random cities held out and tested on Sep-Oct, ridge scores 2.27% and LightGBM 2.89%.

## 7. The judgment call

The win isn't clean. LightGBM beats ridge on Jul-Aug and Sep-Oct. A ridge with no market or time terms also wins Sep-Oct, 1.86% vs 2.24%, and that's the fold most like November-December. I still chose the final model because its worst fold is 3.00% against 5.67%, and it keeps the weekly pattern that the simpler version can't produce. If rates fall back in November-December, the simpler model would likely have done better.

## 8. December chart




The weekly wave, about $823 to $848, comes from the daily market values. The training data has no holiday season, so the model has no holiday effect, and the high on December 25 is just that week's Thursday peak.

## 9. Limitations

With only one year of data, I can't separate a long-term trend from seasonality. I expect about 2-3% MAPE on clean rows, based on the folds. If Spotter's labels contain the same 1.4% corruption, dollar errors will be much higher, with MAE around $105-120. I also chose the model using only a few folds, so small differences between models aren't reliable.